



Stanford

A Space Debris Consciousness Manifesto:

Bridging Public Awareness, Policy, and Technology for Orbital Sustainability

Iyesomi Eigbe

MS, Department of Aeronautics & Astronautics

Stanford University | somisash@stanford.edu

Abstract

Orbital debris is a critical, yet publicly "invisible," threat to global infrastructure. This research proposes a novel, three-phase socio-technical framework to build a global culture of "Space Debris Consciousness." By integrating gamified education, policy mobilization, and AI-driven autonomous systems, the framework creates a scalable pathway from public awareness to autonomous action, directly addressing the cultural and political gaps that hinder current efforts.

The Core Challenge: A Governance and Awareness Vacuum

Unlike terrestrial environmental issues, orbital debris lacks public visibility and a binding global governance structure. This research tackles this dual challenge head-on, arguing that technological solutions cannot succeed without the cultural will and policy frameworks to support them.

The Integrated Framework

Phase 1: Cultivating Public Urgency and Policy Mobilization

This phase establishes the cultural and policy foundation required for long-term orbital stewardship. It focuses on transforming orbital debris from an abstract technical issue into a public, political, and educational priority.

Key elements include:

- **Gamified Public Engagement:** Using simulations and accessible visual tools to help non-experts intuitively understand conjunctions, debris growth, and collision cascades.
- **Educational Outreach:** Creating curriculum kits and classroom-ready demonstrations to introduce orbital sustainability into high school and undergraduate STEM environments.
- **Policy Momentum:** Drawing on models like the Astra Carta to frame a governance ethos grounded in responsibility, transparency, and long-term thinking. It also explores initiatives to accelerate creative solutions for debris mitigation.

Phase 2: Intelligent Decision-Making for Orbital Safety

Phase 2 provides the cognitive layer that connects policy aspirations to operational capability. The goal is to adapt decision-making frameworks, particularly POMDPs, to enable spacecraft to:

- act under uncertainty,
- reduce false alarms,
- optimize maneuver planning, and
- improve coordination across satellite operators.

These AI tools will also serve as educational demonstrations within Phase 1 engagement platforms, showing how autonomous systems "think" during conjunction scenarios.

Phase 3: Autonomous Mitigation (Long-Term Vision)

The final phase maps the pathway toward active remediation: moving from avoiding debris to systematically removing it.

This includes:

- conceptual architecture for autonomous capture, relocation, or deorbiting systems,
- integration of Phase 2 decision models into robotic platforms, and
- long-term alignment with global sustainability frameworks to ensure orbital hygiene becomes a standard expectation, not an aspirational ideal.

Phase 3 positions the framework as a continuum, where early cultural awareness leads to informed policy, which enables AI-supported safety, which ultimately supports autonomous stewardship.

Methodology & Next Steps

As this is the first quarter of a multi-quarter research program, the work is now focused on concept development and early design across three parallel tracks that map onto the phases of the framework.

Track A: Public Engagement Tools

In this initial stage, I am developing the concept and early prototype for a web-based Conjunction Decision Game designed to teach the fundamentals of orbital risk and decision-making. Alongside this, I am outlining curriculum kits for high school and undergraduate students to introduce core concepts of orbital sustainability in an accessible way.

Track B: Technical Modeling and Simulation

I am designing the initial architecture for an orbital simulation environment and specifying how POMDP-based models will be used to explore how AI systems can support autonomous debris avoidance and safe satellite operations. During the next quarter, the focus is on defining the problem formulation, data and scenario assumptions, and model architecture, with implementation and quantitative benchmarking to follow in future work.

Track C: Leadership Engagement and Knowledge Archiving

This track grounds the framework in proven strategies by engaging visionary leaders and preserving their insights. I have begun outreach to key figures whose early advocacy shaped public understanding of complex sustainability issues, including a formal request to His Majesty King Charles III, architect of the Astra Carta. As part of this work, I am developing *The Space Debris Revolution Series*, an archival project to document the insights of pioneers in sustainability and stewardship. These outreach efforts will directly inform the public engagement and policy dimensions of the framework. A parallel goal is to develop the blueprint for a global innovation prize, similar to the Earthshot Prize, to accelerate breakthroughs in debris removal and responsible satellite operations.

Finally, to further strengthen the policy dimensions of this framework in future quarters, I am enrolling in the Winter/Spring Policy Lab on governing autonomous and AI systems in outer space (AA262/810G), which will inform the ongoing development of this research project.

This research is conducted under the advising of Dr. Sigrid Elschot in the Department of Aeronautics & Astronautics at Stanford University for the AA290 research course.